THIRTEEN

The Discussion Section

It's in the Discussion that writing practice varies the most, not just from discipline to discipline but even from paper to paper. The Discussion thus offers you great freedom in content and organization. For some writers, this freedom is exhilarating; for others, it's terrifying. For me, it's both.

Your Discussion's function is to turn data into knowledge. That means harnessing raw results to answer your paper's central research question(s) and to relate your answers to your broader discipline. The Discussion considers and extends your results to claim the strongest interpretation and the broadest importance that you can legitimately argue. All parts of that last sentence are important. It's entirely appropriate (and necessary) to blow your own horn by pointing out how your results provide novel understanding, resolve a longstanding dispute, or challenge or overturn previous knowledge—presuming, of course, that they do. It's also entirely appropriate to step beyond firm inference and speculate about what your results *might* mean. At the same time, reviewers and readers will notice, and resent, attempts to oversell results, to gloss over limitations in the data, or to misrepresent speculation as inference. Experienced writers avoid overstepping these rhetorical bounds with what linguists call "hedges" (Hyland 1998). Hedges are words or phrases that limit or fine-tune the strength of a claim: for instance, "It is *likely* that . . . ," "An *alternative interpretation* is that . . . ," "These data *suggest* . . . ," or "*If correct*, our model explains . . ." Of course, in order to hedge, you need to be aware of limitations or uncertainties in your argument. This means seeing your work as if you were a critical reviewer. This can be difficult (chapter 21), so it is very helpful, before tackling the

Discussion, to present your work at lab meetings or conferences where you can hope to be asked critical questions.

However it looks, your Discussion will be closely tied to your Introduction. The two are complementary: every question raised in the Introduction should be answered in the Discussion, and every major issue treated in the Discussion should be signalled in the Introduction.

The Elements of a Discussion

Although there is no universal recipe for constructing a Discussion, we can recognize four common components: interpretation of results, consideration of weaknesses, broader implications, and future prospects. These usually appear in that order—although some papers omit the weaknesses and/or prospects components—and they represent a gradual broadening of the paper's focus from the waist of the hourglass (Figure 13.1) out to its base. There's a lot of variation in the emphasis given the different components and in the rhetorical moves used to accomplish them (Swales 1990, 2004; Peacock 2002; Basturkmen 2012). Taking the four components in turn, and returning to our star-formation example, we can explore some common ingredients and some good practices:

- **Interpret results to answer your research question**. Because nearly all readers will attach importance to the first paragraph of a Discussion, this is the perfect place to supply a concise answer to your re-

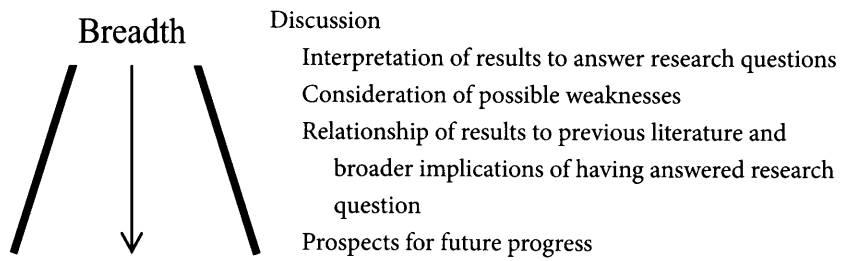

Figure 13.1 Structure and components of a typical Discussion. The Discussion is shown in the context of the whole paper's structure in Figure 8.1.

search question. For instance, either the first or the last sentence of our short opening paragraph might be "Our results imply that the cluster-assist mechanism is not necessary to explain massive-star formation in the Orion, Eagle, and Carina Nebulae."

Your Discussion will typically restate some of your results. But this does not mean simple repetition; use brief reminders and summarize rather than repeat in detail: "Solitary massive protostars were quite common in all three nebulae." The idea is to return the reader's attention to patterns pointed out in the Results, while considering what those patterns mean and how they answer the research question. It can be useful to complement this coarser-level summary of the results with specific examples that support explanations: "Furthermore, the most massive protostar observed (V3416) was one of the most isolated objects in our survey." However, remember that specific examples are just data points, and overdoing this means arguing from anecdote rather than analysis. Answering the research question may also draw on comparisons or synthesis of your current results with results from the literature.

A common rhetorical move is to comment on whether your results were expected or unexpected. In the latter case, you will usually identify what your expectation was, and where it came from: "Surprisingly, among massive protostars with close neighbors, the most massive did not have closer neighbors (on average) than moderately massive ones." Keep in mind, though, that surprises are likely to leave your readers unsatisfied unless you can also offer some explanation. The more support your explanation can have, the better—but it needn't be definitive and sometimes the best you can do is for it to be plausible. We might follow our protostar-surprise sentence with "This suggests that if cluster-assist does happen, it may only require a threshold level of gravitational assist to trigger a process that is subsequently independent of the neighboring masses. Theoretical models have not yet considered this possibility."

- **Consider possible weaknesses**. While there's likely to be hedging throughout the Discussion, it's often useful to include a paragraph or short subsection that explicitly addresses limitations to your inference. For instance, you might consider statistical power (which determines how the reader should interpret the *absence* of a looked-for pattern), weaker steps in a chain of logic, gaps in your data, or alternatives to your preferred interpretation of the results. We might, for instance, point out that our three nebulae aren't fully representative of all star-forming regions: "Our data leave open the possibility that the cluster-assist mechanism is more important in nebulae with higher gas densities; current instrumentation is unable to image any such nebulae with sufficient resolution."

Early-career writers sometimes greatly overdo this Discussion component, perhaps because we emphasize it when we teach undergraduates to write lab reports. If you leave your readers with the impression that your results don't mean much, they will feel they've wasted their time reading the manuscript.

- **Relate your results to previous literature, and considering broader implications**. While you may have called on data from the literature to help answer your precise research question (the first component), in this case I mean fitting that answer into its broader literature context. How does the answer to your research question relate to other research questions (answered or unanswered) in your discipline? Does it support current hypotheses in your field, or call them into question? Does it open up new questions? Does putting your work together with work from the literature establish consensus, draw attention to conflict, or advance our understanding in ways that individual papers can't? For instance, our star-formation results might be extended this way: "The relatively high frequency of solitary massive protostars complements the results of †Nasrallah et al. (2011)[1], who showed a high incidence of radiation jets from condensing protostars. Together, these lines of evidence suggest that the cluster-assist model is not necessary to explain massive protostar formation. They favor instead alternative models in which massive protostars form when gas clouds are heterogeneous in density, leading to asymmetric collapse and radiation escape in localized jets." With support from the literature, your paper's hourglass can regain the width it had at the beginning of the Introduction. Your ultimate aim is to show how the answer to your research question matters to the broadest possible set of readers.

[1] This reference is a fictional example, and throughout this book, I indicate such examples by the dagger (†) prefix.

- **Consider prospects for future progress.** Just as your work has inevitably built on past work by you and your colleagues, your results are likely to suggest directions for future work. For example, perhaps you have studied a specific case and your results point to a path to more general understanding: "While massive-star formation in the Orion, Eagle, and Carina Nebulae seems not to depend on the cluster-assist mechanism, future work should consider whether this conclusion can be generalized to nebulae differing in gas density and morphology. Planned enhancements to ALMA's resolution should soon make our approach feasible for a very large set of star-forming nebulae." Note that you should not simply offer the platitude that more research is necessary; instead, indicate specific directions such research should take. It may seem as if you're selling your own work short by admitting that it doesn't answer all questions. Not at all! In fact, you can emphasize the importance of your work in part by showing how it identifies future research directions that are likely to be profitable.

Do You Need a Conclusion?

One final issue bedevils writers of Discussions. Should your Discussion have a Conclusion? Of course you should have a small-c "conclusion": the answer to your research question. The question is whether this calls for a big-C "Conclusion" *section* (either a headed subsection at the end of the Discussion, or a separate section following it). There's no consensus on this question: practices vary among and within disciplines.

My own feeling is that separate Conclusion sections are seldom needed. When they exist, such sections usually do three things: reiterate the research question and its answer; state the significance or application of that answer; and identify directions for future research. But these are already typical elements of the Discussion. Unless your Discussion is very long and complex (or poorly organized), these points should be fresh in the reader's mind at its end. Because the last paragraph of your Discussion (like the first) is a power position–something readers will pay special attention to—this is a good place for a brief reminder of your most important finding and its significance. Extensive repetition to build a formal Conclusion section, however, is likely redundant. If your

paper does seem to need a separate Conclusion section, before writing one, think about whether this is a sign that your Discussion needs to be simplified or reorganized. If your paper actually is complex enough that your reader can profit from a separate Conclusion, keep it short (perhaps four to six sentences) and avoid simply repeating the Discussion.

Chapter Summary

- The Discussion is the least formulaic of the major sections of a paper.
- The Discussion considers results and places them in context to claim the strongest interpretation and broadest importance you can legitimately argue.
- The Discussion complements the Introduction, dealing with every major issue raised there—but only those.
- Most Discussions (1) interpret results to answer the research question; (2) consider possible weaknesses of the study; (3) relate results to previous literature to suggest broad implications; and (4) consider prospects for future progress.
- Whether or not the Discussion ends with an explicitly labelled Conclusion, it should end with a brief reminder of your most important finding and its importance.

Exercises

1. Choose a recently published paper in your field, and read the Discussion. Highlight text comprising each of the four main Discussion components (interpreting results, considering weaknesses, relating results to literature, considering future progress).
2. What occupies the power position at the end of the Discussion you chose? What else might the author have placed there, and how might that choice have changed the paper?